Track 1: AI for Accounting SW 1

AI-Based Fixed Asset Depreciation Prediction

Features Report

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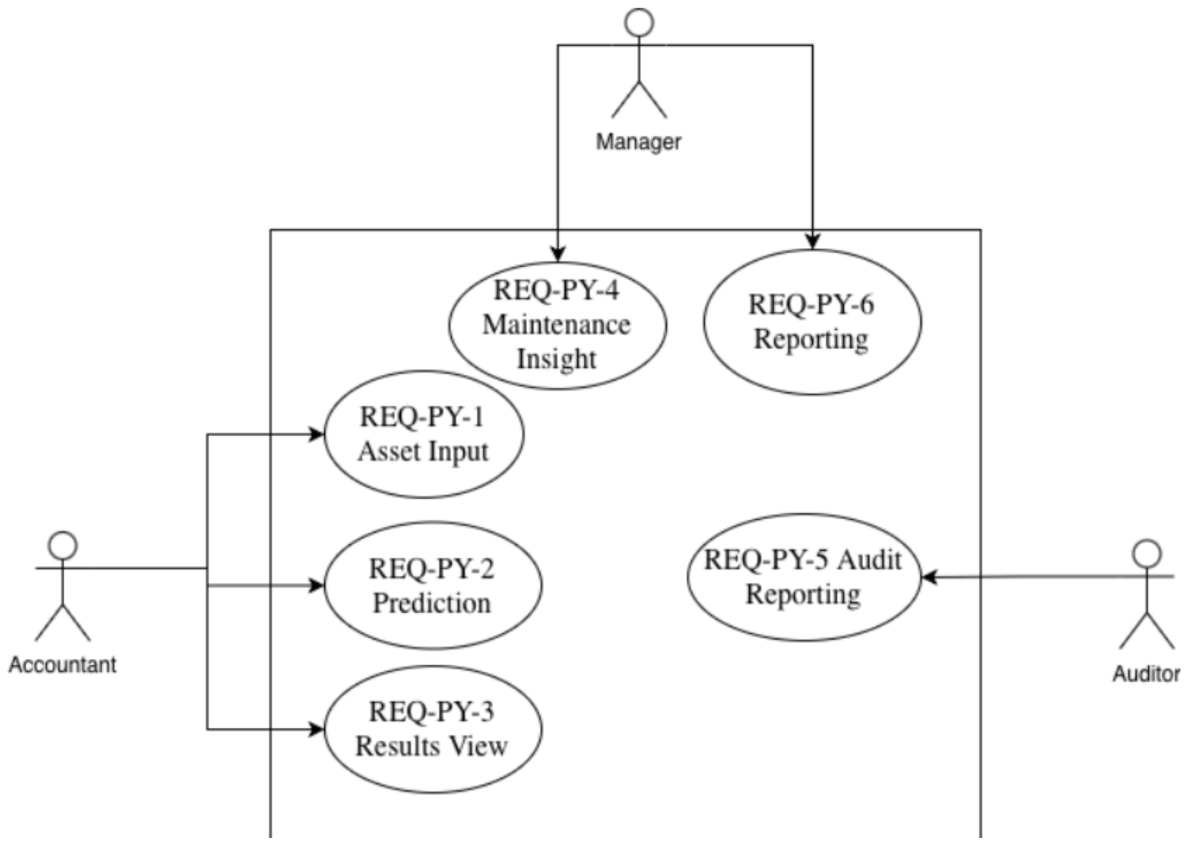

Figure 1: Improved Use Case Diagram

This figure proves the main use case of the proposed fixed asset depreciation prediction feature and shows the interaction between the Accountant, Manager, Auditor, and the system.

Project Features

1. AI-Based Fixed Asset Depreciation Prediction

This project feature focuses on applying artificial intelligence in fixed asset management by designing a prototype for depreciation prediction. The prototype uses a regression-based machine learning concept to support the prediction of asset value, depreciation amount, and estimated maintenance cost.

The feature is designed for accounting and business users. It allows the accountant to enter asset-related data such as cost, age, usage level, maintenance information, and asset type. The system then conceptually processes these inputs and presents prediction outputs that can support fixed asset management decisions.

This report includes the improved design artifacts after expert feedback. Therefore, the improved use case diagram, improved depreciation prediction BPMN, improved user stories, and improved test cases are used in place of the original versions.

1.1. Business Process 1

The proposed feature includes two main business processes. The first process is the improved depreciation prediction process, where the user enters asset data, the system validates the input, and the regression-based machine learning concept generates depreciation-related outputs together with asset insights.

- Improved Depreciation Prediction Process: collects asset data, validates inputs, applies the regression-based prediction concept, generates depreciation and maintenance outputs, and creates asset insights.

The second process is the report generation process, where the user requests a report after prediction results are available and the system prepares the report based on the generated outputs.

- Report Generation Process: retrieves prediction results, generates a report, and presents it to the user.

1.1.1. Use Cases

The improved use case diagram shows the main actors and their interactions with the proposed system after expert feedback. The main actors are the Accountant, Manager, and Auditor. The Accountant is related to asset input, prediction, and result viewing. The Manager is related to maintenance insight and reporting. The Auditor is limited to audit reporting functionality.

Use: Draw.io

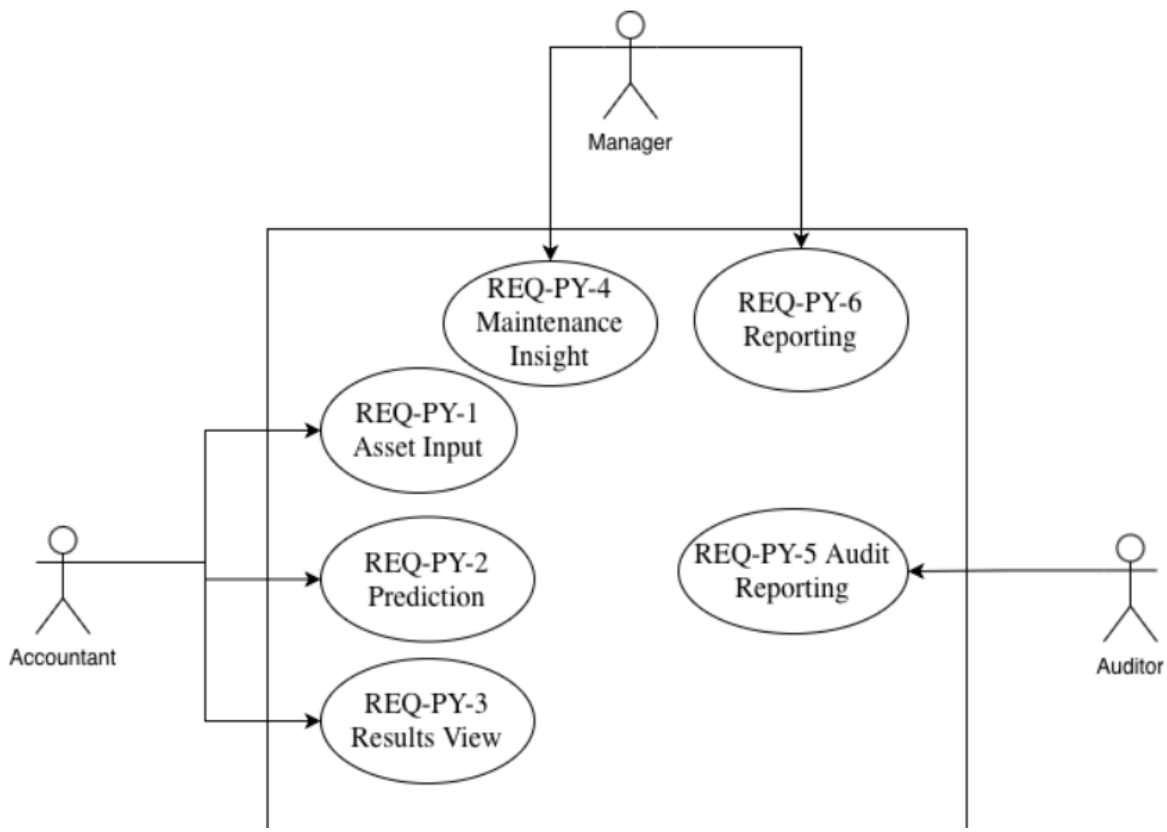

Figure 2: Improved Use Case Diagram

1.1.2. Business Process Diagram (BPMN)

The improved depreciation prediction BPMN helps visualize the conceptual framework of the prediction process, the process begins when the user enters the asset data, the system then validates the input data based on business rules. If the data is invalid, the system displays a validation error. If the data is valid, the system processes the data using the regression-based machine learning concept, generates the depreciation prediction, estimates maintenance cost, generates asset insights, and displays the prediction results.

Use: https://camunda.com/

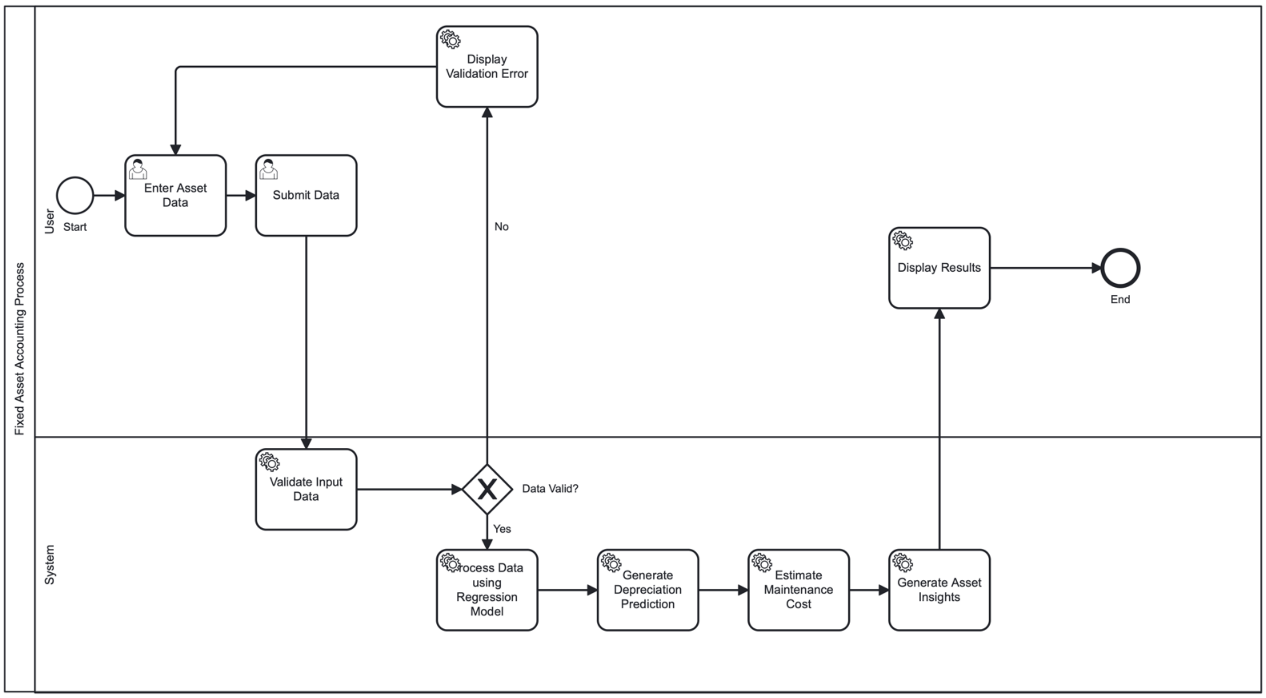

Figure 3: Improved Depreciation Prediction BPMN

The report BPMN shows how a report is generated after prediction results are available. The user requests a report, the system retrieves the prediction data, generates the report, and displays it to the user.

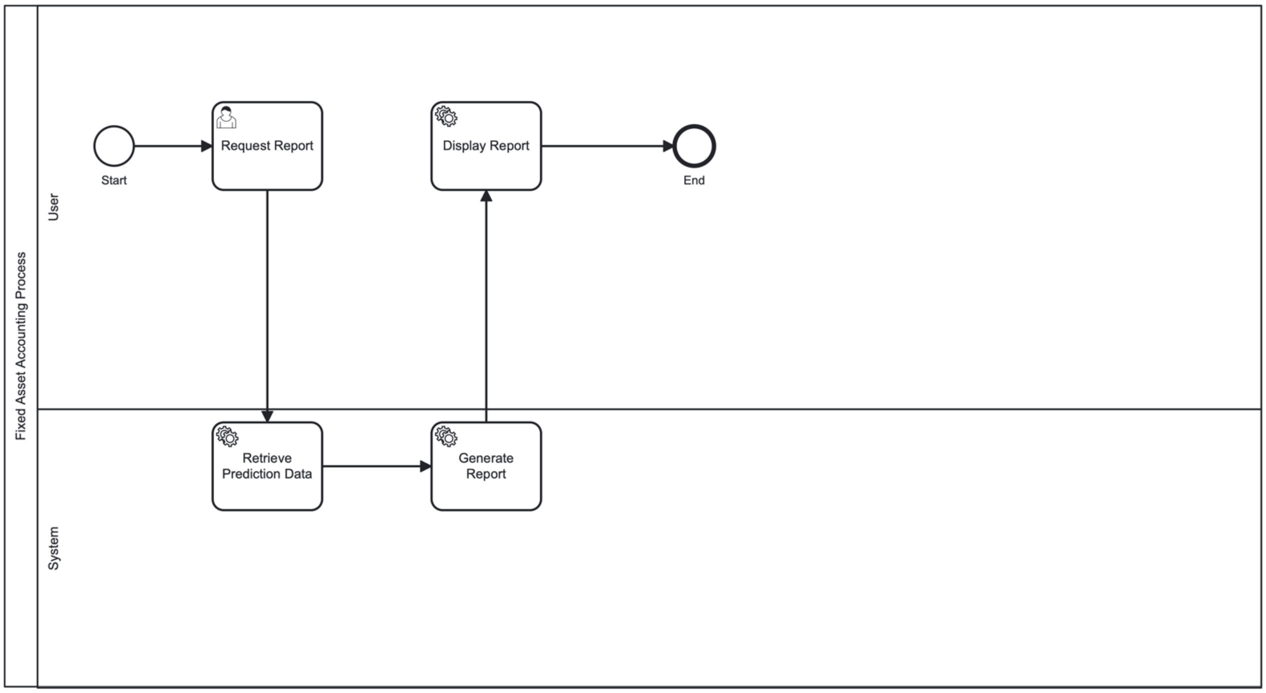

Figure 4: Reporting BPMN

1.1.3. Improved User Stories

Table 1 presents the improved user stories of the proposed system, including the requirement ID, epic, and user story summary for each main system function after expert feedback.

Requirement ID	Epic	User Story Summary
REQ-PY-1	Asset Input	As an Accountant, I want to enter asset data (cost, age, usage, maintenance, asset type) so that the system can process it for prediction.
REQ-PY-2	Prediction	As an Accountant, I want the system to generate predicted asset value and depreciation so that I can analyze asset performance.
REQ-PY-3	Results View	As an Accountant, I want to view the predicted results clearly so that I can understand the output.
REQ-PY-4	Maintenance Insight	As a Manager, I want to see estimated maintenance cost and asset insights so that I can evaluate asset efficiency.
REQ-PY-5	Audit Reporting	As an Auditor, I want to review the generated reports so that I can use the prediction results for auditing purposes.
REQ-PY-6	Reporting	As a Manager, I want to generate reports so that I can use the results for decision-making.

Table 1: Improved User Stories

Field Requirements / Business Rules

Table 2 presents the field requirements and business rules that control the input data, output values, validation logic, asset insight generation, and reporting behavior of the prototype.

Field / Output	Requirement Type	Requirement / Business Rule
Asset Cost	Field Requirement	The asset cost must be entered as a number greater than zero. The system should not accept zero or negative asset cost.
Asset Age	Field Requirement	The asset age must be entered as a number greater than or equal to zero. Negative age values are not accepted.
Usage Level	Field Requirement	The usage level must be selected from predefined options such as low, medium, or high in order to keep the input standardized.
Maintenance Information	Field Requirement	The maintenance field must be completed before starting the depreciation prediction process. It can represent the maintenance condition or maintenance cost depending on the prototype design.
Asset Type	Field Requirement	The asset type must be selected from predefined categories such as equipment, machinery, vehicle, or building. This helps the system process asset data in a consistent format.
Mandatory Fields	Business Rule	All required fields must be completed before the user can run the depreciation prediction process. Missing mandatory fields should lead to a validation message.
Predicted Asset Value	Business Rule	The predicted asset value must not be negative and should not be higher than the original asset cost. This keeps the output within a logical accounting range.
Depreciation Amount	Business Rule	The depreciation amount must not be negative and must not exceed the original asset cost. This prevents illogical depreciation results.
Estimated Maintenance Cost	Business Rule	The estimated maintenance cost must be displayed as zero or a positive value. It should not appear as a negative amount.
System Feedback	Business Rule	The system should provide feedback when the entered data is valid or invalid, so the user can understand whether the prediction can be processed.
Asset Condition Insight	Business Rule	The asset condition insight should explain the meaning of the prediction results in a simple way, such as whether the asset is moderately depreciated or has average operational efficiency.
Report Generation	Business Rule	A report should only be generated after prediction results are available. The report should depend on the generated depreciation and maintenance outputs.

Table 2: Field Requirements and Business Rules

Wireframe

The prototype wireframes were created using Figma. They demonstrate how users interact with the proposed system through an input screen and an improved output screen based on expert feedback.

Asset Depreciation Prediction Input Screen: https://www.figma.com/make/Gy6C0U1dVPS9ddjf6diuYZ/Fixed-Asset-Input-Screen?p=f&t=hcASyYvTi1HgVMyq-0

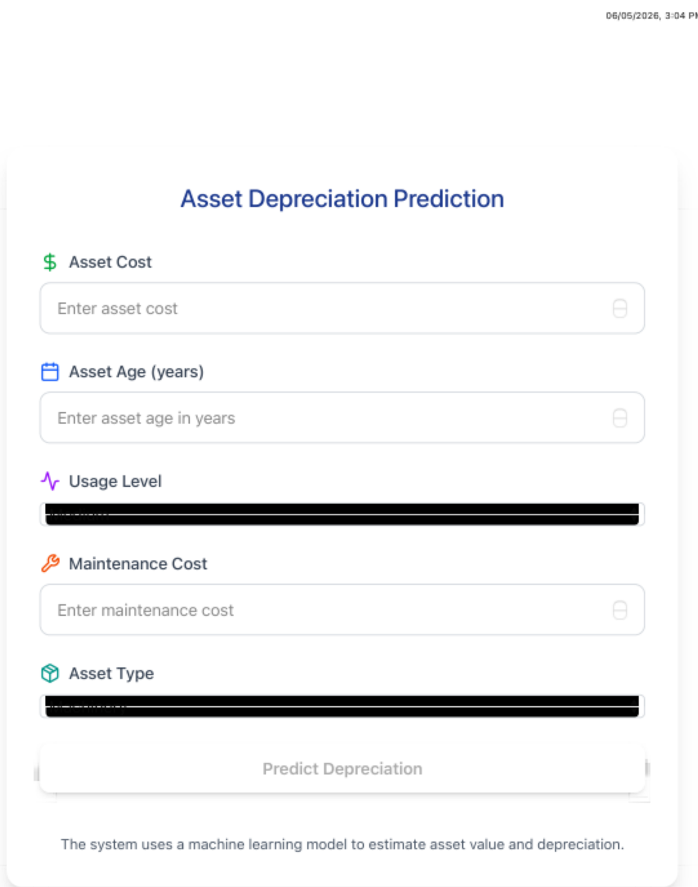

Figure 5: Asset Depreciation Prediction Input Screen

Improved Depreciation Prediction Results Output Screen:
https://www.figma.com/make/Ks3hPctNsbo5J3x8jyNMHi/Depreciation-Prediction-Results-Screen?t=YVzqYi8ai8rXJ2HR-0

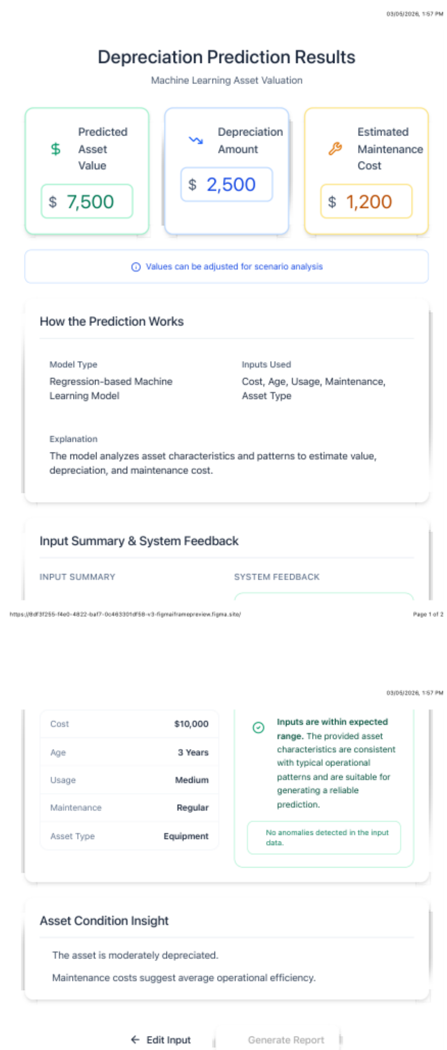

Figure 6: Improved Depreciation Prediction Results Output Screen

Improved Test Cases

Table 3 presents the improved test cases used to check whether the prototype functions correctly after expert feedback, including valid input, prediction generation, result display, reporting, maintenance insight, and validation scenarios.

Test Case ID	User Story	Test Scenario	Test Steps	Expected Result
TC1	REQ-PY-1	Valid asset data input	1) Enter asset cost. 2) Enter age. 3) Select usage and maintenance. 4) Submit data.	System accepts valid input data successfully.
TC2	REQ-PY-2	Generate depreciation prediction	1) Enter valid asset data. 2) Submit prediction request.	System displays predicted depreciation amount and asset value.
TC3	REQ-PY-3	Display prediction results	1) Submit valid asset data. 2) Open results screen.	System displays prediction outputs clearly.
TC4	REQ-PY-4	Generate maintenance insight	1) Enter maintenance information. 2) Submit data.	System displays estimated maintenance cost and asset insights.
TC5	REQ-PY-5	Generate audit report	1) Generate prediction results. 2) Request report generation.	System generates report for auditing purposes successfully.
TC6	REQ-PY-6	Generate management report	1) Generate prediction results. 2) Generate report.	System generates report for decision-making purposes.
TC7	REQ-PY-1	Invalid negative asset cost	1) Enter negative asset cost. 2) Submit data.	System displays validation error message.
TC8	REQ-PY-1	Invalid negative asset age	1) Enter negative asset age. 2) Submit data.	System prevents prediction and displays validation feedback.
TC9	REQ-PY-1	Missing mandatory fields	1) Leave required fields empty. 2) Submit data.	System requests completion of missing information before prediction.

Table 3: Improved Test Cases

Additional Notes:

No real dataset was used in this project. The machine learning approach is presented conceptually as a regression-based prototype and was not trained or tested using real fixed asset data also no software installation is required to review the project because the implementation is prototype-based.